

# Experiment No. 03

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**Subject: DE**

**Class: TE H-2**

## Title

**Relational Schema Design, SQL Normalization and PostgreSQL Implementation**

## Aim

To design a relational database for a University Management System, normalize it up to Third Normal Form (3NF), implement the schema using PostgreSQL, populate the database with sample data, perform analytical SQL queries, apply window functions, and optimize query performance using indexes and execution plans.

## Objectives

- To understand relational database design.
- To convert a problem statement into an ER model.
- To normalize relations up to Third Normal Form (3NF).
- To implement normalized tables in PostgreSQL.
- To define Primary Keys and Foreign Keys.
- To insert and manage relational data using SQL.
- To perform analytical SQL queries using JOIN, GROUP BY, HAVING, and subqueries.
- To implement SQL Window Functions.
- To analyze query execution using EXPLAIN and EXPLAIN ANALYZE.
- To understand query optimization using indexes.

# Theory

A **Relational Database Management System (RDBMS)** stores data in the form of related tables. Each table consists of rows and columns, where relationships between tables are maintained using Primary Keys and Foreign Keys.

Database normalization is the process of organizing data to minimize redundancy and improve consistency. In this experiment, the database is normalized up to **Third Normal Form (3NF)**.

- **First Normal Form (1NF):** Removes repeating groups and ensures atomic values.
- **Second Normal Form (2NF):** Eliminates partial dependency on composite keys.
- **Third Normal Form (3NF):** Eliminates transitive dependency and ensures that every non-key attribute depends only on the primary key.

The University Management System consists of the following entities:

- Department
- Faculty
- Student
- Course
- Semester
- Enrollment

These entities are related through Primary Keys and Foreign Keys, ensuring referential integrity.

The schema is implemented using PostgreSQL Data Definition Language (DDL) commands such as CREATE DATABASE and CREATE TABLE.

After creating the schema, sample data is inserted using SQL INSERT statements. Data is then retrieved and analyzed using SQL queries including:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- JOIN
- Aggregate Functions
- Subqueries

Advanced SQL Window Functions such as **ROW\_NUMBER()**, **RANK()**, **DENSE\_RANK()**, **LAG()**, **LEAD()**, **Running Total**, and **Running Average** are used to generate analytical reports while preserving row-level information.

Query optimization techniques such as **Indexes**, **EXPLAIN**, and **EXPLAIN ANALYZE** are applied to improve query execution performance.

Finally, OLAP-style analytical SQL queries are executed to generate reports including department-wise performance, merit lists, semester analysis, and grade distribution.

## Software Requirements

- Ubuntu Linux
- PostgreSQL 17
- psql Terminal
- Nano Editor

## Database Used

**Database Name:** university\_db

## Tables Created

1. Department
2. Faculty
3. Student
4. Semester
5. Course
6. Enrollment

## SQL Concepts Used

- CREATE DATABASE
- CREATE TABLE

- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- CHECK
- INSERT
- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- INNER JOIN
- Aggregate Functions
- Subqueries
- Window Functions
- Indexing
- EXPLAIN
- EXPLAIN ANALYZE

## Window Functions Used

- ROW\_NUMBER()
- RANK()
- DENSE\_RANK()
- LAG()
- LEAD()
- SUM() OVER()
- AVG() OVER()

## Query Optimization Techniques

- Single Column Index
- Composite Index
- EXPLAIN
- EXPLAIN ANALYZE

# Result

The relational database for the University Management System was successfully designed, normalized up to Third Normal Form (3NF), implemented in PostgreSQL, populated with sample data, and verified. SQL queries, analytical functions, window functions, and query optimization techniques were executed successfully. The desired outputs were obtained.

```
Percentage : 88%
=====
Report saved at /home/dataeng/reports/Exp3_124A8113_20260717_054405.txt
dataeng@IDS-LABH-06:~$ ls
DE_Exp2_Evaluator.sh datasets downloads projects reports software venvs
dataeng@IDS-LABH-06:~$ cat /home/dataeng/reports/Exp3_124A8113_20260717_054405.txt
=====
DATA ENGINEERING LAB AUTO EVALUATION REPORT
=====
Student : Rohit Swami
Roll : 124A8113
Batch : W-2
Faculty : Akshay Jadhav
DB : university_db

System Verification

Database Verification

Row Counts
department : 3
faculty : 9
student : 20
semester : 2
course : 10
enrollment : 36

Constraints

Database Objects
tables : 70
views : 141
sequences : 6
indexes : 11
functions : 0
triggers : 0

SQL Verification

Window Functions

Optimization

Mini Challenge

=====
Score : 37 / 42
Percentage : 88%
dataeng@IDS-LABH-06:~$ |

Activate Windows
Go to Settings to activate Windows.
```

```
Percentage : 88%
=====
Report saved at /home/dataeng/reports/Exp3_124A8114_20260717_054405.txt
dataeng@IDS-LABH-06:~$ ls
DE_Exp2_Evaluator.sh datasets downloads projects reports software venvs
dataeng@IDS-LABH-06:~$ cat /home/dataeng/reports/Exp3_124A8114_20260717_054405.txt
=====
DATA ENGINEERING LAB AUTO EVALUATION REPORT
=====
Student : Shrivani Talekar
Roll : 124A8114
Batch : W-2
Faculty : Akshay Jadhav
DB : university_db

System Verification

Database Verification

Row Counts
department : 3
faculty : 9
student : 20
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SQL Verification

Window Functions

Optimization

Mini Challenge

=====
Score : 37 / 42
Percentage : 88%
dataeng@IDS-LABH-06:~$ |

Activate Windows
Go to Settings to activate Windows.
```

```
dataeng@AIDS-LAB4-BE: ~
(2,3,1,79,'B'),
(3,1,1,84,'A'),
(4,4,1,73,'B'),
(5,2,2,95,'A');
INSERT 0 6
university_db=# SELECT * FROM department;
 department_id | department_name
-----
1 | Computer Engineering
2 | Information Technology
3 | Electronics Engineering
(3 rows)

university_db=# SELECT * FROM faculty;
 faculty_id | faculty_name | email | department_id
-----
1 | Dr. Sharma | sharma@university.edu | 1
2 | Prof. Patel | patel@university.edu | 2
3 | Dr. Mehta | mehta@university.edu | 3
(3 rows)

university_db=# SELECT * FROM student;
 student_id | first_name | last_name | email | phone | dob | department_id
-----
1 | Rahul | Patil | rahul@university.edu | 9876543219 | 2003-02-15 | 1
2 | Sneha | Joshi | sneha@university.edu | 9876543211 | 2002-11-20 | 2
3 | Amit | Kulkarni | amit@university.edu | 9876543212 | 2003-04-18 | 1
4 | Priya | Deshmukh | priya@university.edu | 9876543213 | 2002-09-11 | 3
5 | Rohan | Shinde | rohan@university.edu | 9876543214 | 2003-01-05 | 2
(5 rows)

university_db=# SELECT * FROM course;
 course_id | course_name | credits | faculty_id | department_id
-----
1 | Database Systems | 4 | 1 | 1
2 | Operating Systems | 4 | 1 | 1
3 | Web Technologies | 3 | 2 | 2
4 | Digital Electronics | 4 | 3 | 3
(4 rows)

university_db=# SELECT * FROM semester;
 semester_id | semester_name | academic_year
-----
1 | Semester I | 2025-26
2 | Semester II | 2025-26
(2 rows)

university_db=# SELECT * FROM enrollment;
 enrollment_id | student_id | course_id | semester_id | marks | grade
-----
1 | 1 | 1 | 1 | 88.00 | A
```

Activate Windows  
Go to Settings to activate Windows.

## Conclusion

The experiment demonstrated the complete lifecycle of relational database development, including schema design, normalization, implementation, data manipulation, analytical SQL, window functions, and query optimization. The use of PostgreSQL ensured data integrity through constraints and efficient query processing using indexing and execution plans. The experiment provided practical knowledge of relational database management and analytical SQL techniques used in modern data engineering applications.